# Power Budget: User Interface Board

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|---|---|
| Team Number: | 103 |
| Project Name: | GripRx |
| Team Member Names: | Duotao Gao, Gabriel Toneser Facchin, Zice Sun |
| Version: | 5 (Duotao Gao, 2026-10-23) |

**A. List all major powered components, excluding power sources, regulators and passive elements.**

| All Major Components | Component Name | Part Number | Supply Voltage Range | # | Budget current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | PIC18F57Q43 Curiosity Nano (MCU board) | DM164150 | 1.8 - 5.5 V | 1 | 50.0 | 50.0 | mA |
| | Character LCD + backlight | NHD-0216K1Z-FSW-FBW-L | 4.5 - 5.5 V | 1 | 23.0 | 23.0 | mA |
| | Quad op amp + speaker drive | MCP6004-I/P | 1.8 - 6.0 V | 1 | 20.0 | 20.0 | mA |

**B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design. Add additional power rails or change the power rail voltages if needed.**

| +12V Power Rail (not used) | Component Name | Part Number | Supply Voltage Range | # | Budget current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | None on this board | — | — | 0 | 0.0 | 0.0 | mA |
| | | | | | **Subtotal** | 0.0 | mA |
| | | | | | **Safety Margin** | 25% | |
| | | | | | **Total Current Required on +12V** | 0.0 | mA |
| Regulator or Source Choice | Not used (no +12 V loads) | — | — | 0 | 0.0 | 0.0 | mA |
| | | | | | **Total Remaining Current Available on +12V Rail** | 0.0 | mA |

| +5V Power Rail | Component Name | Part Number | Supply Voltage Range | # | Budget current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | PIC18F57Q43 Curiosity Nano (MCU board) | DM164150 | 1.8 - 5.5 V | 1 | 50.0 | 50.0 | mA |
| | Character LCD + backlight | NHD-0216K1Z-FSW-FBW-L | 4.5 - 5.5 V | 1 | 23.0 | 23.0 | mA |
| | Quad op amp + speaker drive | MCP6004-I/P | 1.8 - 6.0 V | 1 | 20.0 | 20.0 | mA |
| | Other small circuit loads | LEDs, buttons, pots, bias | 5 V budget | 1 | 13.3 | 13.3 | mA |
| | No other +5 V loads | - | - | 0 | 0.0 | 0.0 | mA |
| | | | | | **Subtotal** | 106.3 | mA |
| | | | | | **Safety Margin** | 25% | |
| | | | | | **Total Current Required on +5V Rail** | 132.8 | mA |
| c2. Regulator or Source Choice | +5V linear regulator | L7805ABV-DG | 7.5 - 18 V | 1 | 1500.0 | 1500.0 | mA |
| | | | | | **Total Remaining Current Available on +5V Rail** | 1367.2 | mA |

| -5V Power Rail (not used) | Component Name | Part Number | Supply Voltage Range | # | Budget current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | None: all op amps are single-supply | — | — | 0 | 0.0 | 0.0 | mA |
| | | | | | **Subtotal** | 0.0 | mA |
| | | | | | **Safety Margin** | 25% | |
| | | | | | **Total Current Required on -5V** | 0.0 | mA |
| Regulator or Source Choice | Not used (no negative rail) | — | — | 0 | 0.0 | 0.0 | mA |
| | | | | | **Total Remaining Current Available on -5V Rail** | 0.0 | mA |

| +3.3V Power Rail (not used) | Component Name | Part Number | Supply Voltage Range | # | Budget current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | None on this board | — | — | 0 | 0.0 | 0.0 | mA |
| | | | | | **Subtotal** | 0.0 | mA |
| | | | | | **Safety Margin** | 25% | |
| | | | | | **Total Current Required on +3.3V** | 0.0 | mA |
| Regulator or Source Choice | Not used (no +3.3 V loads) | — | — | 0 | 0.0 | 0.0 | mA |
| | | | | | **Total Remaining Current Available on +3.3V Rail** | 0.0 | mA |

**C. Select a regulator or source for each rail. Confirm that the total remaining current on every rail is not negative.**

**D. Select an external source and confirm that it can supply all connected regulators simultaneously.**

| External Power Source 1 | Component Name | Part Number | Supply Voltage Range | V | Rated current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| Power Source 1 Selection | 9 V 3 A plug-in wall adapter, issued in class | BestCH 9 V 3 A (B09ZTKTLGW) | See adapter label | 9 | 3000.0 | 3000.0 | mA |
| Power Rails Connected to External Source 1 | Regulator ground-current allowance | L7805ABV-DG | 9 V input | 1 | 6.0 | 6.0 | mA |
| | +5V linear regulator | L7805ABV-DG | 7.5 - 18 V | 1 | 1500.0 | 1500.0 | mA |
| | +3.3V regulator (not used) | — | — | 0 | 0.0 | 0.0 | mA |
| | **Total Remaining Current Available on External Power Source 1** | | | | | 1494.0 | mA |

| External Power Source 2 (not used) | Component Name | Part Number | Supply Voltage Range | V | Rated current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| Power Source 2 Selection | Not used (single 9 V adapter) | — | — | — | 0.0 | 0.0 | mA |
| Power Rails Connected to External Source 2 | -5V regulator (not used) | — | — | 0 | 0.0 | 0.0 | mA |
| | **Total Remaining Current Available on External Power Source 2** | | | | | 0.0 | mA |

**E. Calculate battery life, if applicable, using the battery capacity and simultaneous system demand.**

| | Component Name | Part Number | Supply Voltage Range | Capacity (mAh) | Required by Regulators | |
|---|---|---|---|---|---|---|
| | No battery (wall-powered board) | — | — | N/A | 0.0 | |

Battery Life: N/A (wall-powered board)

**Notes**

All powered parts use +5 V. The unused rails and second source are marked not used. No battery is included.

Current basis: Nano 50 mA and MCP6004 with speaker drive 20 mA are engineering allocations. LCD maximum is 23 mA. Auxiliary loads are 13.25 mA. These allocations require confirmation with the final firmware and prototype.

The audio output uses MCP6004 U3A, C14 = 220 uF and R16 = 330 ohm. A 2.5 V excursion gives at most about 7.6 mA through R16 alone. Speaker current is included once in the 20 mA allocation.

The +5 V budget is 132.81 mA including 25% margin. The source check reserves 1.5 A regulator capacity plus 6 mA ground current. Estimated regulator dissipation is about 0.59 W. Confirm input voltage, current, cooling and audible output.

References: Newhaven NHD-0216K1Z-FSW-FBW-L specification, p. 5; Microchip DS20001733L; ST L78 datasheet, Table 3. Full links are on the webpage.